Physics Lectures

Quantum Field Theory

Lecture Notes

Quantization, Fock space, propagators, scattering, and renormalization.

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Contents

Preface: What These Notes Are For

> **Remark**
>
> These notes are designed as a continuation of the Classical Field Theory I–II sequence. They do not try to re-teach variational calculus, Lorentz covariance, Noether currents, gauge redundancy, or the classical equations of fundamental fields from the beginning. Instead, they start where the classical notes stop: with fields as dynamical systems ready to be quantized.

Quantum Field Theory After Classical Field Theory

Classical field theory teaches us to describe nature by local fields, actions, symmetries, conservation laws, and equations of motion. Quantum field theory keeps this architecture, but changes the meaning of the basic objects. A classical field configuration is no longer the final description of a physical state. It becomes an ingredient in a quantum theory whose observable content is encoded in states, operators, correlation functions, transition amplitudes, and probabilities.

The guiding idea of the first part of these notes is simple: a free classical field decomposes into normal modes, and each normal mode behaves like a harmonic oscillator. Quantizing the field therefore means quantizing infinitely many oscillators at once. The excitations of these oscillators are interpreted as particles.

What Will Not Be Repeated

The notes assume familiarity with the following ideas from classical field theory:

- the action principle and Euler–Lagrange equations for fields,
- canonical momenta and Hamiltonian density,
- Lorentz covariance and relativistic notation,
- real and complex scalar fields,
- electromagnetic and Yang–Mills gauge fields at the classical level,
- Noether's theorem and conserved currents,
- Green's functions as solutions of linear differential equations.

Whenever one of these topics is used in a quantum calculation, the notes recall just enough of the classical structure to make the calculation readable. The aim is not repetition, but conversion: classical structures are translated into quantum structures.

What Is New in QFT

The genuinely new concepts are the vacuum, Fock space, field operators, propagators, Wick contractions, Feynman diagrams, scattering amplitudes, loop corrections, regularization, renormalization, and scale dependence. These ideas are not decorative additions to classical field theory. They are the mechanisms that make quantum field theory predictive.

A central conceptual warning will appear repeatedly: a Feynman diagram is not a literal picture of tiny particles following tiny trajectories. It is a compact bookkeeping device for terms in a perturbative expansion of correlation functions or scattering amplitudes.

How to Read These Notes

A useful rhythm for reading a QFT calculation is:

$$\text{classical field} \longrightarrow \text{mode expansion} \longrightarrow \text{operators} \longrightarrow \text{states} \longrightarrow \text{correlators} \longrightarrow \text{amplitudes.}$$

At first this may look like a change of notation. It is not. Each arrow changes what is meant by a physical prediction. The purpose of the notes is to make those changes explicit and technically usable.

Part I

Quantization of Fields

Chapter 1

From Classical Normal Modes to Quantum Oscillators

Remark

The first bridge from classical field theory to quantum field theory is the normal-mode decomposition of a free field. A free field is not quantized as one object at once. It is decomposed into independent oscillating modes, and each mode is quantized as a harmonic oscillator.

1.1 Starting Point: A Classical Field Decomposed into Modes

Consider a real classical field in a box of volume V, with boundary conditions chosen so that the allowed spatial momenta are discrete. A typical free-field solution can be expanded as a sum of normal modes,

$$\phi(t, \mathbf{x}) = \sum_{\mathbf{k}} q_{\mathbf{k}}(t) u_{\mathbf{k}}(\mathbf{x}),$$

where $u_{\mathbf{k}}(\mathbf{x})$ are spatial mode functions and $q_{\mathbf{k}}(t)$ are time-dependent mode amplitudes.

For a free theory, the modes decouple. After inserting the mode expansion into the classical action, the field theory becomes a sum of independent mechanical systems,

$$L = \sum_{\mathbf{k}} \left(\frac{1}{2}\dot{q}_{\mathbf{k}}^2 - \frac{1}{2}\omega_{\mathbf{k}}^2 q_{\mathbf{k}}^2 \right),$$

up to conventional normalization factors. The frequency is determined by the field equation. For a relativistic scalar field of mass m,

$$\omega_{\mathbf{k}} = \sqrt{\mathbf{k}^2 + m^2}$$

in units with $c = \hbar = 1$.

1.2 Each Mode as a Harmonic Oscillator

The expression above is the Lagrangian of a collection of independent harmonic oscillators. Each momentum label $\mathbf{k}$ names one oscillator. The canonical momentum conjugate to $q_{\mathbf{k}}$ is

$$p_{\mathbf{k}} = \dot{q}_{\mathbf{k}},$$

and the Hamiltonian takes the schematic form

$$H = \sum_{\mathbf{k}} \left(\frac{1}{2}p_{\mathbf{k}}^2 + \frac{1}{2}\omega_{\mathbf{k}}^2 q_{\mathbf{k}}^2 \right).$$

Example

The field is infinite-dimensional, but the free field is not mysterious at this stage. It is a very large, eventually infinite, collection of uncoupled harmonic oscillators. The conceptual leap is that we quantize all of these oscillators and reinterpret their excitations as particles.

1.3 Quantizing a Single Oscillator

Before quantizing the field, recall the single harmonic oscillator. Its canonical variables q and p satisfy the commutation relation

$$[\hat{q}, \hat{p}] = i.$$

The Hamiltonian is

$$\hat{H} = \frac{1}{2}\hat{p}^2 + \frac{1}{2}\omega^2\hat{q}^2.$$

It is useful to introduce creation and annihilation operators,

$$\hat{a} = \sqrt{\frac{\omega}{2}}\,\hat{q} + \frac{i}{\sqrt{2\omega}}\,\hat{p}, \qquad \hat{a}^\dagger = \sqrt{\frac{\omega}{2}}\,\hat{q} - \frac{i}{\sqrt{2\omega}}\,\hat{p}.$$

They satisfy

$$[\hat{a}, \hat{a}^\dagger] = 1.$$

In terms of them, the Hamiltonian becomes

$$\hat{H} = \omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right).$$

1.4 Creation and Annihilation Operators

The operator $\hat{a}^\dagger$ raises the oscillator excitation number, while $\hat{a}$ lowers it. If $|0\rangle$ is the oscillator vacuum, then

$$\hat{a}|0\rangle = 0,$$

and the normalized n-th excited state is obtained by applying $\hat{a}^\dagger$ repeatedly,

$$|n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}|0\rangle.$$

The number operator is

$$\hat{N} = \hat{a}^\dagger\hat{a},$$

with eigenvalues $0, 1, 2, \ldots$.

1.5 From One Oscillator to Infinitely Many Oscillators

For a field, every mode receives its own ladder operators,

$$\hat{a}_{\mathbf{k}}, \qquad \hat{a}^\dagger_{\mathbf{k}}.$$

The canonical commutation relations become

$$[\hat{a}_{\mathbf{k}}, \hat{a}^\dagger_{\mathbf{k}'}] = \delta_{\mathbf{k}\mathbf{k}'}, \qquad [\hat{a}_{\mathbf{k}}, \hat{a}_{\mathbf{k}'}] = 0, \qquad [\hat{a}^\dagger_{\mathbf{k}}, \hat{a}^\dagger_{\mathbf{k}'}] = 0.$$

The vacuum of the free field is the state annihilated by all annihilation operators,

$$\hat{a}_{\mathbf{k}}|0\rangle = 0 \qquad \text{for all } \mathbf{k}.$$

A one-particle state of momentum $\mathbf{k}$ is

$$|\mathbf{k}\rangle = \hat{a}^\dagger_{\mathbf{k}}|0\rangle.$$

A many-particle state is obtained by applying several creation operators.

Definition

The *Fock space* of a free quantum field is the Hilbert space generated from the vacuum by applying creation operators for all allowed modes. It is the natural Hilbert space for states with variable particle number.

1.6 Field Quanta as Mode Excitations

The word "particle" in free quantum field theory means an excitation of a field mode. This is a precise statement in the free theory: the Hamiltonian is a sum of number operators,

$$\hat{H} = \sum_{\mathbf{k}} \omega_{\mathbf{k}} \left(\hat{a}^\dagger_{\mathbf{k}} \hat{a}_{\mathbf{k}} + \frac{1}{2} \right).$$

The state $\hat{a}^\dagger_{\mathbf{k}}|0\rangle$ has one quantum of energy $\omega_{\mathbf{k}}$ above the vacuum, and momentum $\mathbf{k}$.

This interpretation becomes more subtle in interacting theories. Interactions mix modes, dress particles, shift masses, generate unstable resonances, and make the vacuum itself nontrivial. Nevertheless, the free-field construction is the starting point for perturbative QFT.

1.7 What Is New Compared with Classical Field Theory?

Classical field theory assigns numbers to field configurations. Quantum field theory assigns operators to fields and vectors to states. The basic object is no longer a definite field profile $\phi(x)$, but an operator $\hat{\phi}(x)$ acting on a Hilbert space.

The same classical equation may still appear, but with a different role. For a free scalar field, the Klein–Gordon equation becomes an equation satisfied by the field operator,

$$(\Box + m^2)\hat{\phi}(x) = 0.$$

This does not mean that the field has one definite classical value. It means that the operator-valued distribution evolves according to the free field equation.

Remark

The first lesson of QFT is therefore not that particles are little balls. It is that particle states are organized excitations of quantum fields. The oscillator algebra is the mechanism that turns a classical mode into a quantum excitation.

Appendix A

Quantization Dictionary

Remark

This appendix records the translation rules that are used repeatedly in the main text. They should not be treated as mechanical replacements valid in every context. They are starting points that must be checked against constraints, symmetries, domains of operators, and gauge redundancies.

A.1 Classical Mode to Quantum Oscillator

A classical normal mode with amplitude $q_{\mathbf{k}}(t)$ and frequency $\omega_{\mathbf{k}}$ is quantized like a harmonic oscillator. Schematically,

$$q_{\mathbf{k}}, p_{\mathbf{k}} \quad \longrightarrow \quad \hat{q}_{\mathbf{k}}, \hat{p}_{\mathbf{k}}, \qquad [\hat{q}_{\mathbf{k}}, \hat{p}_{\mathbf{k}'}] = i\delta_{\mathbf{k}\mathbf{k}'}.$$

The oscillator is then rewritten in terms of ladder operators $\hat{a}_{\mathbf{k}}$ and $\hat{a}_{\mathbf{k}}^{\dagger}$.

A.2 Poisson Bracket to Commutator

The canonical rule is

$$\{A, B\}_{\text{PB}} \quad \longrightarrow \quad \frac{1}{i}[\hat{A}, \hat{B}].$$

For fields, the equal-time canonical relation is

$$[\hat{\phi}(t, \mathbf{x}), \hat{\pi}(t, \mathbf{y})] = i\delta^{(3)}(\mathbf{x} - \mathbf{y}).$$

A.3 Classical Field to Field Operator

A classical field $\phi(x)$ becomes an operator-valued distribution $\hat{\phi}(x)$. The word “distribution” is important: products of field operators at the same spacetime point are generally singular and require a careful definition.

A.4 Classical Green’s Function to Propagator

A classical Green’s function solves a differential equation with a delta-function source. A quantum propagator is a vacuum expectation value of a time-ordered product,

$$\Delta_F(x - y) = \langle 0 | T\{\hat{\phi}(x)\hat{\phi}(y)\} | 0 \rangle.$$

It is related to Green’s functions, but its physical meaning is quantum correlation, not merely classical signal propagation.

A.5 Classical Symmetry to Quantum Ward Identity

A classical continuous symmetry gives a conserved current. In the quantum theory, the same symmetry constrains correlation functions and scattering amplitudes through Ward identities. If the symmetry fails after quantization, the failure is called an anomaly.

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